

Full DSA Guide

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I. Core Data Structures

1. Linear Data Structures

- 1.1 Arrays - Fundamental contiguous memory structure
- 1.2 Strings - Sequences of characters, often with specialized algorithms
- 1.3 Linked Lists
 - 1.3.1 Singly Linked List
 - 1.3.2 Doubly Linked List
 - 1.3.3 Circular Linked List
- 1.4 Stack - Last-In, First-Out (LIFO) structure
- 1.5 Queue - First-In, First-Out (FIFO) structure
 - 1.5.1 Simple Queue
 - 1.5.2 Circular Queue
 - 1.5.3 Deque (Double-Ended Queue)

2. Non-Linear Data Structures

- 2.1 Trees
 - 2.1.1 Binary Tree (Pre-order, In-order, Post-order, Level-order)
 - 2.1.2 Binary Search Tree (BST)
 - 2.1.3 AVL Tree (Self-Balancing BST)
 - 2.1.4 Red-Black Tree
- 2.2 Graphs
 - 2.2.1 Representation: Adjacency List & Adjacency Matrix
 - 2.2.2 Types: Directed, Undirected, Weighted, Unweighted

3. Hash-Based Structures

- 3.1 Hash Table / Hash Map - Key-value pair storage
- 3.2 Set - Stores unique elements

4. Advanced Data Structures

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- 4.1 Heap / Priority Queue
- 4.2 Trie (Prefix Tree)
- 4.3 Union-Find (Disjoint Set Union - DSU)
- 4.4 Segment Tree
- 4.5 Fenwick Tree (Binary Indexed Tree - BIT)
- 4.6 Bloom Filter

II. Core Algorithms & Techniques

5. Searching

- 5.1 Linear Search
- 5.2 Binary Search (and its variations)
- 5.3 Jump Search
- 5.4 Exponential Search

6. Sorting

6.1 Comparison-Based

- 6.1.1 Bubble Sort
- 6.1.2 Selection Sort
- 6.1.3 Insertion Sort
- 6.1.4 Merge Sort
- 6.1.5 Quick Sort
- 6.1.6 Heap Sort
- 6.1.7 Timsort

6.2 Non-Comparison-Based

- 6.2.1 Counting Sort
- 6.2.2 Radix Sort
- 6.2.3 Bucket Sort

7. Algorithmic Paradigms

7.1 Recursion & Backtracking

- 7.1.1 Factorial, Fibonacci
- 7.1.2 Subsets, Permutations

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7.1.3 N-Queens, Sudoku

7.1.4 Maze Problems

7.2 Divide and Conquer

7.2.1 Merge Sort

7.2.2 Quick Sort

7.2.3 Binary Search

7.2.4 Closest Pair of Points

7.3 Greedy Algorithms

7.3.1 Fractional Knapsack

7.3.2 Activity Selection Problem

7.3.3 Huffman Coding

7.4 Dynamic Programming (DP)

7.4.1 0/1 Knapsack

7.4.2 Longest Increasing Subsequence (LIS)

7.4.3 Longest Common Subsequence (LCS)

7.4.4 Matrix Chain Multiplication

7.4.5 Coin Change Problem

7.4.6 DP on Trees

7.4.7 Bitmask DP

7.5 Kadane's Algorithm

8. Common Techniques

8.1 Two Pointers

8.2 Sliding Window

8.3 Monotonic Stack / Queue

III. Graph & Tree Algorithms

9. Graph Traversal

9.1 Breadth-First Search (BFS)

9.2 Depth-First Search (DFS)

10. Shortest Path Algorithms

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10.1 Dijkstra's Algorithm

10.2 Bellman-Ford Algorithm

10.3 Floyd-Warshall Algorithm

10.4 Johnson's Algorithm

11. Minimum Spanning Tree (MST)

11.1 Prim's Algorithm

11.2 Kruskal's Algorithm

12. Other Graph Algorithms

12.1 Topological Sort

12.2 Cycle Detection

12.3 Tarjan's Algorithm (SCC)

12.4 A* Search Algorithm

12.5 Ford-Fulkerson Algorithm (Max Flow)

13. Tree Algorithms

13.1 Tree Traversals

13.2 Lowest Common Ancestor (LCA)

13.3 BST Operations

13.4 Tree Balancing Techniques

IV. Miscellaneous & Advanced Topics

14. String Algorithms

14.1 Pattern Searching

14.1.1 KMP Algorithm

14.1.2 Rabin-Karp Algorithm

14.1.3 Boyer-Moore Algorithm

14.2 Longest Palindromic Substring

14.3 Manacher's Algorithm

14.4 Suffix Array and Suffix Tree

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15. Mathematical Concepts

15.1 Bit Manipulation

15.2 Number Theory

15.2.1 Greatest Common Divisor (GCD)

15.2.2 Sieve of Eratosthenes

15.3 Simplex Algorithm (Linear Programming)